Adaptive Mid-Level Feature Attention Learning for Fine-Grained Ship Classification in Optical Remote Sensing Images

Xi Yang, *Member, IEEE*, Zilong Zeng, and Dong Yang

***Abstract*— Ship classification in optical remote sensing images is a critical task for various maritime applications, including anti-smuggling, maritime traffic control, and maritime rescue. However, fine-grained ship classification (FGSC) is challenging due to the complex background, intraclass similarity, and interclass difference. In this article, we propose a novel mid-level feature attention learning method for FGSC. Our method incorporates mid-level feature casual attention (MFCA) and mid-level channel attention (MCA) to identify discriminative regions and local features corresponding to subtle visual features. The MFCA constrains the learning process of mid-level features through comparison with attention maps and counterfactual attention maps, while the MCA uses a discriminative component to extract discriminative features from channel information and a diversity component to focus feature channels on more obvious feature regions. Besides, an adaptive weight is added to dynamically adjust the influence of MFCA and MCA in the model. Our method can be trained end-to-end and requires no annotations other than category information. Extensive experiments on two large-scale FGSC datasets, FGSC-23 and FGSCR-42, demonstrate that the proposed method achieves state-of-the-art performance, outperforming existing methods by a significant margin.**

***Index Terms*— Attention mechanism, fine-grained ship classification (FGSC), mid-level feature, remote sensing.**

I. Introduction

SHIPS play a crucial role in various marine activities, serving as a primary tool for ocean transportation and combat. As a result, the automatic positioning and fine-grained classification of ships have become critical tasks. Fine-grained ship classification (FGSC) is essential for providing necessary information for strategic decision-making, shortening the decision-making cycle, and forming the basis for real-time battlefield situation awareness in large-scale combat environments. Previously, synthetic aperture radar (SAR) images were widely used for all-weather monitoring of ship targets, and the technology has matured over time. However, with the continuous development of modern remote sensing technology, there has been a shift toward high-resolution optical remote sensing images, which contain more detailed information. Consequently, ship detection and classification based on optical remote sensing images have become a primary focus of research and development.

Fine-grained classification typically involves two primary approaches: localizing discriminative parts for part-specific feature extraction and exploiting relationships between fine-grained labels to regularize feature learning. While methods using fine-grained labels have reported decent results, it has become apparent that such supervised approaches are not scalable due to the difficulty of obtaining expert human annotations, which are error-prone and time-consuming. To overcome this problem, more recent research [1], [2], [5] has focused on localizing parts using attention mechanisms, which extract fine-grained features independently while ignoring the relationship between features. However, larger patches used in these methods are more sensitive to deformations, misalignments, and parts not falling entirely on a single surface or object, resulting in unreliable decision-making during the training process.

To address these limitations, researchers have proposed integrating mid-level features into the fine-grained classification task. Mid-level features are feature maps extracted before the advanced features, containing less advanced semantic information but richer discriminate feature information. Luo et al. [21] tried to uncover the information contained in mid-level features and proposed a cross-category cross-semantic regularizer to guide the extracted features to represent semantic parts. Rao et al. [22] also provided a tool to measure attention quality and a powerful supervisory signal to guide the learning process. The above methods demonstrate that integrating information from mid-level features into the network's features significantly improves the model's performance in fine-grained classification tasks.

However, applying these approaches to FGSC is challenging due to the complex and dense image content of remote sensing images. For instance, Zhang et al. [38] proposed an attribute-guided multilevel enhanced feature representation network (AMEFRN), and although its final predicted labels were correct on FGSC-23, the visual decision process of the network did not entirely focus on the discriminative region of the objects. As shown in Fig. 1(a), attention methods classify images mainly based on both foreground and background information. These spurious associations between attention regions and predicted labels result in unreliable decision-making in the training process. The attention methods only focus on the discriminative patterns and extract invariant deep

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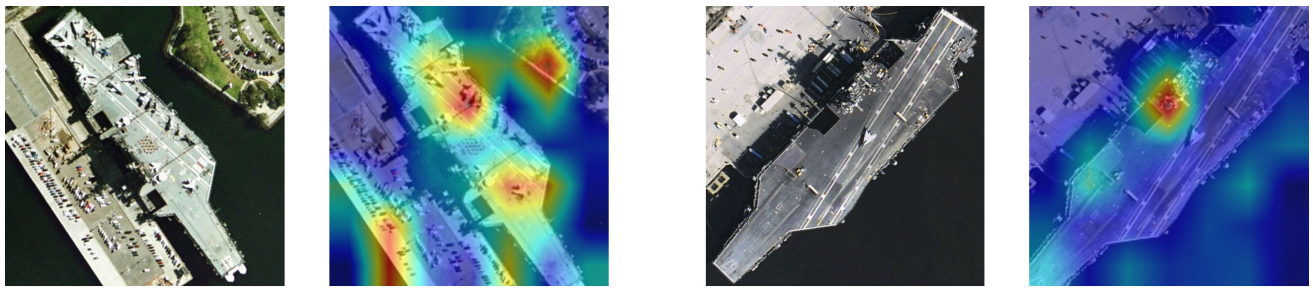


Fig. 1. Images of aircraft carrier from the FGSCR-42 dataset and its corresponding activation maps. (a) Attention algorithms classify images mainly based on background information. (b) Attention algorithms only focus on salient features.

semantic features. Besides, as shown in Fig. 1(b), the learning process without fine-grained labels encourages the model to only focus on some salient features instead of distinguishing features, which causes the attention method to concentrate on suboptimal features. Therefore, it is necessary to introduce more discriminate feature information to help the attention model focus on subtle discriminative features.

In this article, we propose a mid-level feature counterfactual attention to mine the causal relationship between mid-level features and class labels. Mid-level feature casual attention (MFCA) enables the model to focus on both main and suboptimal features, reducing the weight of salient features in classification. We also propose mid-level channel attention (MCA) to obtain more diverse discriminative features. MCA allows multiple channels of mid-level features to represent each category through loss function constraints, providing the network with an independent feature learning direction for each category. Specifically, we propose counterfactual attention (CA) learning to mine the causal relationship between mid-level features and labels and then introduce MCA to obtain diverse attention regions. Our method introduces minimal additional computational cost during training while significantly improving the fine-grained learning capabilities of attention models for mid-level features.

This article makes several contributions to the field of FGSC. Specifically, the following conditions hold.

1) We propose an MFCA to extract fine-grained feature information of mid-level features in the spatial dimension. MFCA explores the relationships between mid-level features and category labels from different images and different layers to learn discriminative fine-grained features. MFCA enables the network to extract more meaningful features and achieve better classification performance.
2) We propose an MCA to extract diverse features of mid-level features in the channel dimension. MCA makes the feature channels of mid-level features corresponding to each category and extracts diverse and fine-grained features between subcategories in the channel dimension.
3) We propose an adaptive loss (AL), which can adaptively change the weights of MFCA and MCA in the cross-entropy loss according to the classification confidence of the high-level features. AL is beneficial to accelerate network convergence and balance the influence of high-level features and mid-level features in the training process.

II. Related Work

A. Fine-Grained Visual Classification

Recent research [3], [9], [17], [30], [33], [34] has explored more general attention architectures that allow networks to discover and localize discriminative features in images automatically without any additional supervisory signal, making fine-grained classification easier to implement in real-world applications. Sermanet et al. [23] added attention mechanisms to fine-grained classification networks, and this idea was further extended in [26] and [39] who achieved advanced results at the time, illustrating the effectiveness of attention module for fine-grained classification tasks. Yang et al. [35] proposed a self-supervised multiagent cooperative learning scheme to identify discriminative features. Meanwhile, Long et al. [20] innovatively proposed selective sparse sampling networks (S3Ns) to extract discriminative features without losing contextual information. Wang et al. [32] proposed a low-rank representation mechanism called Gaussian mixture model (GMM) to accurately select discriminative details and filter irrelevant information, which solved the inaccurate localization of recognition areas due to the operation of superimposing local receptive areas.

To better mine the interaction relationships of features, some methods encode high-order convolution variables of feature maps to obtain a more compact overall representation. Lin et al. [15] used two side-by-side feature extraction convolutional neural networks (CNNs) to perform an outer product on the features of each position on the image, followed by maximum pooling to obtain the result. However, this approach results in a high feature dimension, leading to an excessive amount of parameters and calculations. Zheng et al. [40] proposed a deep bilinear transformation network (DBT-NET) that is grouped by design so that the channels of the same semantics are pulled close, while the channels of different semantics are pushed away as much as possible. Such paired grouping enables bilinear operations to better capture the details of different semantic groups while also achieving low-dimensional requirements, enabling better performance of bilinear features in fine-grained classification.

B. Fine-Grained Ship Classification

FGSC has become an increasingly important topic in recent years, as advancements in optical remote sensing image technology have made it possible to achieve more accurate and detailed ship classification.

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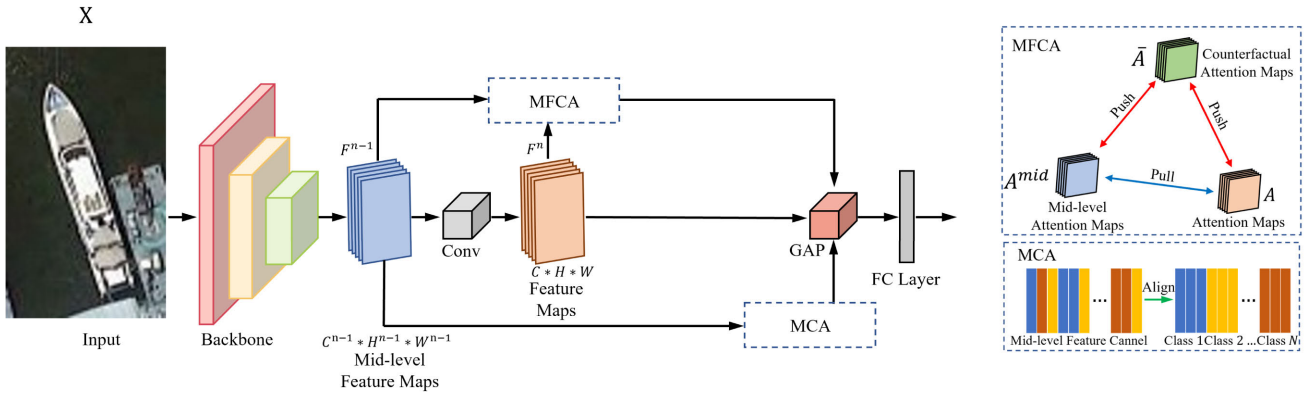


Fig. 2. Architecture of the proposed network.

Limited by the low resolution of optical remote sensing images in the early days, some studies [8], [42] attempted to classify ship and nonship objects. Common remote sensing ship images often increase the difficulty of classification due to factors such as cloud and fog occlusion. To solve this problem, Liu et al. [16] proposed improved Inceptionv3 and center loss CNN (IICL-CNN), which improve the recognition accuracy of fuzzy targets. An objective function was added not only to learn the common features of clear samples and occluded samples but also to reduce the intraclass gap between occluded samples.

A dataset that has been widely used for fine-grained ship detection is the HRSC2016 dataset [19], which contains uniform horizontal boxes and three-level class labels. This dataset has also inspired new baselines for different tasks, including binary classification labels and masks for segmentation tasks. Shi et al. [24] proposed a new ship classification framework multifeature ensemble based on a CNN (ME-CNN). They use multiorder amplitude and phase images as the input of CNN and Gabor filter to obtain global rotation invariant features to make up for the shortcomings of CNN. Moreover, completed local binary patterns (CLBPs) were used to extract local texture information.

In addition, Zhang et al. [37] proposed a dataset namely FGSC-23 consisting of 23 fine-grained categories of ships, which provided several attribute information such as ship aspect ratio in addition to category labels. An AMEFRN is proposed to pay more attention to the silent region and suppress other regions by fusing reweighted regional features. The recent release of the FGSCR-42 dataset [4] has also enabled more extensive research into FGSC. This dataset contains 42 categories and is currently the largest public remote sensing image dataset available for this task.

Although previous studies have focused on high-level feature information, the proposed method in this study introduces a mid-level feature attention network for FGSC in remote sensing images. This method aims to incorporate more detailed texture information for improved classification accuracy.

C. Mid-Level Features

FGSC benefits from incorporating more specific low-level information from mid-level features. Previous works have demonstrated the advantages of improving mid-level convolutional feature representation in fine-grained classification [31]. Interaction between features from multiple layers has also been explored, such as the bottom-up and top-down approaches [14]. Inspired by the above research, Luo et al. [21] proposed a one-squeeze multiexcitation (OSME) module to extract mid-level features and fuse them with high-level features.

While these previous works have mined fine-grained information by training more features at different levels, they ignore the discriminative feature information contained in mid-level features. Simply adding mid-level features and high-level features leads to confusion in high-level semantic information contained in the features. To more effectively utilize mid-level features, we propose a fine-grained attention classification network based on mid-level features. MFCA mines the causal relationship between mid-level features and labels and effectively suppresses salient features. Meanwhile, MCA diversifies the features that channels focus on. By introducing these attention mechanisms, the network better utilizes the detailed information contained in mid-level features for improved FGSC.

III. Methodology

The overall architecture of the proposed method is shown in Fig. 2, and we employ a deep CNN to extract features from input images X, which is then divided into two stages. In the first stage, the high-level features are selected, and the corresponding counterfactual feature maps and attention feature maps are generated. Simultaneously, mid-level features are extracted from the network. In the second stage, the classification results obtained by the proposed method based on CA maps and high-level feature attention maps are compared with those of mid-level features, and the corresponding loss scores are calculated. These loss scores serve as a constraint to guide the mid-level features to focus on regions full of high-level semantic information while avoiding suboptimal solutions caused by salient features. To further diversify the mid-level feature areas that the network focuses on, MCA diversifies the feature channels that the network focuses on at the channel level. These two modules extract the fine-grained information contained in mid-level features from multiple aspects while preserving the semantic information of high-level features.

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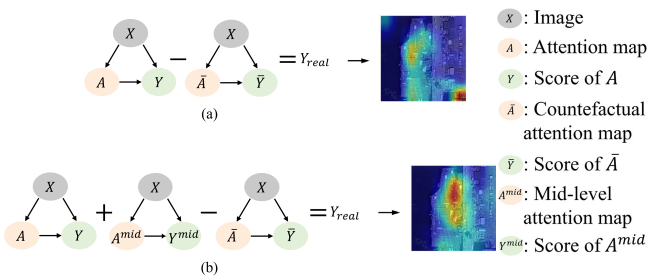


Fig. 3. Difference between CA and MFCA. (a) Causal graph and visualization of CA. (b) Causal graph and visualization of MFCA.

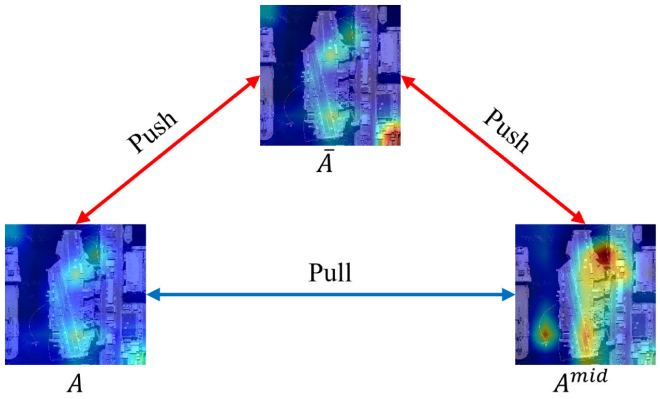


Fig. 4. MFCA designed in terms of causality operates on feature maps.

A. Mid-Level Feature Casual Attention

The MFCA is an indispensable component in analyzing the causal relationships between variables and exploring the impact of attention learning on prediction during the learning process. Traditional CNN models attempt to fit a function to the relationship between data and results based on observational data. However, this approach overlooks the causal relationships [29] between the observed objects and the final prediction outcomes. Therefore, analyzing the causal relationships between variables provides a better understanding of the quality of attention and enables us to think outside of the black box.

As shown in Fig. 3, the introduction of causal maps provides a more intuitive means of evaluating the impact of variables on results through human intervention. In our design diagram, artificial values are assigned to the variables A_{mid} and $\overline{A}$ to sever the link between variable X and these two variables, indicating that the variable is no longer causally related to its parent variable X. MFCA is based on causal inference and aims to simulate human decision-making through counterfactual thinking. We randomly assign attention weights to A_{mid} and $\overline{A}$ and then minimize the distance between the result of the mid-level feature attention feature maps and the higher order attention feature maps while maximizing the distance between the result of the CA feature maps and others.

As shown in Fig. 4, MFCA enables the learning of mid-level attention feature maps to enhance the quality of high-level attention for fine-grained features. Meanwhile, the learning of CA feature maps reduces the attention to salient features and proportionately suppresses background attention. Therefore, our causal map consists of high-level feature attention maps

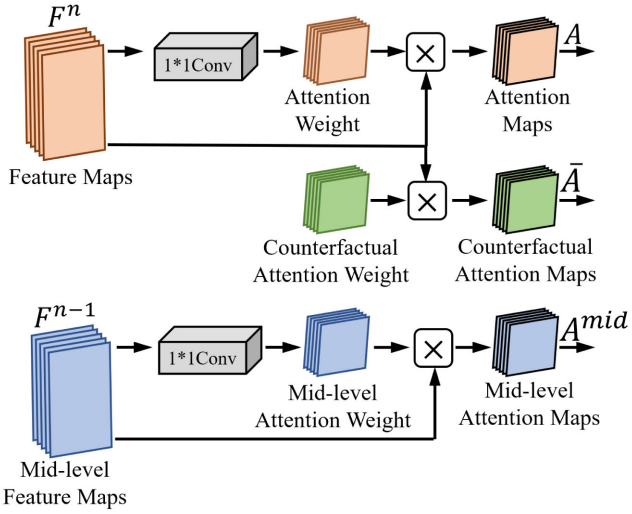


Fig. 5. Framework of the MFCA.

X, CA maps, and mid-level feature maps. Given that the high-level feature maps derive from the mid-level feature maps, the CA maps generated by the high-level feature maps perform a counterfactual intervention on the mid-level feature maps to a certain extent. Specifically, we force the attention algorithms to focus on both salient and suboptimal features, so the attention maps of high-level feature maps will be compared with the attention maps of mid-level maps to focus on salient features from high-level features while learning mid-level features. Then, we guide the attention algorithms to focus on the object instead of the background, so the CA maps will be compared with the attention maps of mid-level maps. Through the subtraction between the two feature scores, randomly generated attention on the background will weaken the ability of mid-level features to pay attention to the background, while randomly generated attention on objects will not weaken much due to the enhancement of high-level features. The specific operation is as follows: first, we predict the result of each attention map to obtain the score of each category, then use this score to perform addition and subtraction, and finally send the score to the loss function.

Based on the constructed causal graph, the proposed MFCA is described in detail in Fig. 5. Given an input image X, high-level features and mid-level features are extracted using CNN. The high-level features are denoted as $F^n \in \mathbb{R}^{C*H*W}$, and the mid-level features are denoted as $F^{n-1} \in \mathbb{R}^{C^{n-1}*H^{n-1}*W^{n-1}}$, where H, W, C, and n represent the height, the weight, and the number of channels and deepest layers in the CNN, respectively. In the MCA, M multihead attention maps are generated by applying a simple convolutional function to the extracted features F^n. These attention maps are represented as $A = \phi(F^n)$, where $\phi(\cdot)$ denotes the convolutional function with a kernel size of 1×1 and the number of attention maps M. The set of attention maps generated is denoted as $A = A_1, A_2, \ldots, A_M$. Notably, $A_M \in A$ represents the attention maps that highlight interesting regions in the input image. The feature maps F^n are weighted by the attention maps generated in the previous step to generate region features r_m after global average pooling. Specifically, the attention maps are applied to the feature maps F^n to obtain a weighted

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representation of the features. The resulting weighted features are then subjected to global average pooling to generate the region features r_m

$$r_m = \phi(F^n \odot A_m) \tag{1}$$

where ϕ represents the global average pooling operation and $\odot$ denotes elementwise multiplication. Then, we obtain the final prediction $Y_{F^n}(A)$

$$Y_{F^n}(A) = fc\left(\sum_{m=1}^{M} r_m\right) = fc\left(\sum_{m=1}^{M} \phi\left(F^n \odot A_m\right)\right) \tag{2}$$

where $fc(\cdot)$ indicates a classifier that includes a fully convolutional layer and a batch normalization layer. Based on the casual inference method, we choose the random attention maps $\overline{A}$ as a CA map. The counterfactual prediction $Y_{F^n}(\overline{A})$ is expressed as

$$Y_{F^n}(\overline{A}) = fc\left(\sum_{m=1}^{M} \overline{r_m}\right) = fc\left(\sum_{m=1}^{M} \phi(F^n \odot \overline{A_m})\right). \tag{3}$$

Let F^{n-1} be the mid-level feature maps obtained from the CNN. These features are fed into a single-layer attention network, which generates mid-level attention maps A^{mid}. The attention map is then used to weight the feature maps F^{n-1} to obtain a weighted representation of the mid-level features. The resulting weighted features are subjected to global average pooling to generate the region features r_m^{mid}. The mid-level region features r_m^{mid} and the mid-level attention maps A^{mid} are combined to produce the final output of the MFCA module. The resulting mid-level attention maps A^{mid}, generated region features r_m^{mid}, and mid-level feature prediction $Y_{F^{n-1}}(A^{\text{mid}})$ are expressed as

$$A^{\text{mid}} = \phi\left(\text{upsample}(F^{n-1})\right) \tag{4}$$

$$r_m^{\text{mid}} = \phi(F^{n-1} \odot A^{\text{mid}}) \tag{5}$$

$$Y_{F^{n-1}}(A^{\text{mid}}) = fc\left(\sum_{m=1}^{M} r_m^{\text{mid}}\right) = fc\left(\sum_{m=1}^{M} \phi(F^{n-1} \odot A_m^{\text{mid}})\right) \tag{6}$$

where the purpose of upsample$(\cdot)$ is to make mid-level features the same size as high-level features.

The real effect of the three parts of attention on the final prediction is represented by the difference between the attention prediction $Y_{F^n}(A)$, its counterfactual prediction $Y_{F^n}(\overline{A})$, and mid-level attention prediction $Y_{F^{n-1}}(A^{\text{mid}})$

$$Y_{\text{real}} = Y_{F^n}(A) + Y_{F^{n-1}}(A^{\text{mid}}) - Y_{F^n}(\overline{A}) \tag{7}$$

where Y_{real} denotes the real effect on prediction.

B. Mid-Level Channel Attention

The extraction of fine-grained features via the MFCA leads to the network assigning an excessive number of feature channels to the same features, which results in a decline in classification performance. In other words, the network only focuses on the main parts of a particular object class, thereby disregarding other crucial features. For example, when classifying an aircraft carrier, the model only focuses on the cockpit, bow, and stern while ignoring features such as the aircraft and runways on the deck. Similarly, when classifying a battleship, the model only focuses on the positions of the main guns and secondary guns, ignoring other relevant features.

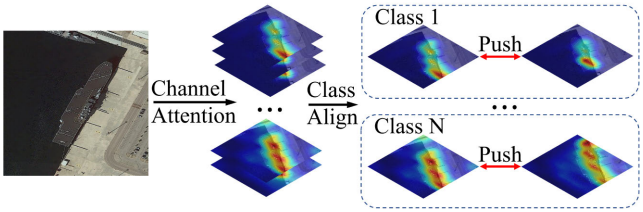


Fig. 6. MCA operates on channels. MCA allows one category to correspond to multiple channels, different colors represent the response of the channel to different categories, and the dark area represents the area that the channel focuses on.

To address these issues, we propose MCA, which facilitates the network's focus on different discriminative regions without requiring fine-grained bounding box or part annotations. MCA consists of two fundamental components: feature channel differentiation and feature channel diversification. The feature channel differentiation component aligns each feature channel with a specific class, ensuring that every feature channel associated with a particular class possesses sufficient distinctiveness. An attention operation is subsequently performed on channel features to compel the network to capture information equally across all channels. Moreover, the random selection of several feature channels as output stimulates each channel to contain discriminative feature information. Randomly selecting channels as output is to make the MCA focus on more different feature channels in a random way and reduce the status of main features. Besides, the random selection of several feature channels as output stimulates each channel to contain discriminative feature information.

The feature channel diversification component aims to direct different feature channels of a category toward distinct regions of the image, thereby preventing the network from assigning an excessive number of feature channels to the same feature and allowing it to focus on more diverse features of the image. By conducting a remote operation between the feature channels of each category, the allocation redundancy of feature channels is reduced by dispersing them, which benefits the network in concentrating on different discriminative features of a category. Specifically, as shown in Fig. 6, MCA aligns each feature channel with each category, and the number of channels generally takes more than twice the number of categories. A loss function is employed to encourage each channel of the network to learn the features of the corresponding category while increasing the distance of each channel.

The framework of MCA is shown in Fig. 7, mid-level features enter the two streams of the network, the left is the discriminative module, which mines discriminative features from channel information, and the right is the diversity module, which focuses the feature channels on more obvious feature regions. The discriminability module loss L_{dis} is represented as

$$L_{\text{dis}}(F^{n-1}) = \text{cross_entropy}\left(\text{softmax}(f(F^{n-1}))\right) \tag{8}$$

$$f\left(F_k^{n-1}\right) = \frac{1}{WH}\sum_{i=1}^{WH} \max_{j=1,2,\ldots,\delta}\left\{M_k \cdot F_{i,j,k}^{n-1}\right\} \tag{9}$$

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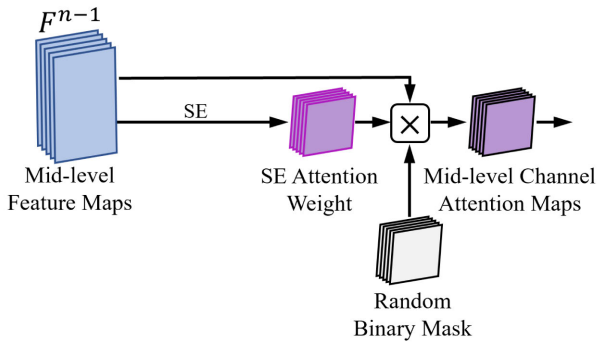


Fig. 7. Framework of the MCA.

where cross_entropy(·) and softmax(·) indicate the cross-entropy loss function and softmax function, respectively, W and H indicate the weight and height of mid-level feature maps, respectively, and M_k is a 0–1 diagonal matrix mask with the same size of F_k^{n-1}.

If the feature channels all focus on the salient features, it will lead to insufficient generalization of the network in identifying fine-grained images, so the feature channels corresponding to a category should focus on different feature areas, to focus on more different areas of the image. The diversity module loss L_{div} is represented as

$$L_{\text{div}}(F^{n-1}) = \frac{1}{c}\sum_{i=0}^{c-1} g\left(F_i^{n-1}\right) \tag{10}$$

$$g(F_i^{n-1}) = \sum_{i=1}^{WH} \max_{j=1,2,\ldots,\delta} \left\{\text{softmax}\left(F_{i,j}^{n-1}\right)\right\} \tag{11}$$

where c indicates the number of channels.

C. Adaptive Loss

Following the processing of the feature maps by the MFCA and MCA, the resulting features are subjected to global average pooling and a fully convolutional layer. The resulting features are then used for classification, and the cross-entropy loss is employed as the classification loss to provide a supervision signal that explicitly guides the learning process of mid-level features. In order to better balance the ratio between losses, we proposed an AL to adaptively adjust the weights according to the high-level feature scores. The total loss function of the MFCA module is represented as

$$L_{\text{normal}} = \text{cross_entropy}(Y, Y_{\text{pred}}) = -\sum_{i=1}^{D} Y^i \log\left(Y_{\text{pred}}^i\right) \tag{12}$$

$$L_{Y_{\text{real}}} = \text{cross_entropy}(Y, Y_{\text{real}}) = -\sum_{i=1}^{D} Y^i \log\left(Y_{\text{real}}^i\right) \tag{13}$$

$$\mu = \lambda * \ln(L_{\text{normal}} + 1) \tag{14}$$

$$\begin{aligned}\text{LOSS} &= L_{\text{normal}} + \mu * (L_{\text{MFCA}} + L_{\text{MCA}}) \\ &= L_{\text{normal}} + \mu * \left(L_{Y_{\text{real}}} + L_{\text{dis}} + L_{\text{div}}\right)\end{aligned} \tag{15}$$

where Y, Y_{pred}, Y_{real}, and D are the ground truth label, the predicted label, the counterfactual predicted label, and the number of categories, respectively. L_{normal} is the cross-entropy loss between the predicted label and the ground truth label, and $L_{Y_{\text{real}}}$ is the cross-entropy loss between the counterfactual predicted label and the ground truth label. μ and λ are adaptive weight and hyperparameter that balance the relative importance of the three losses, respectively. The proposed loss function with L_{normal} aims to adaptively focus on the region of interest and guide the learning of both the MFCA and MCA modules. Notably, since attention loss accounts for the main proportion in the early stage of training and gradually decreases at the end of the training, the network also retains the original loss function to continue to guide the learning of the network. This approach allows the network to benefit from both the original loss function and the proposed loss function, resulting in better performance of the MFCA module.

IV. Experiment

A. Experimental Setup

In this section, we present an evaluation of MFCA, MCA, and AL on two FGSC datasets, FGSC-23 [38] and FGSCR-42 [4]. To ensure consistency with other datasets, we divide both datasets into training and test sets only. Due to the large number of samples in the datasets, we conduct simple data augmentation techniques, including mixup [36], random cropping, and rotation, to supplement image samples of some categories in the training set. We adopt ResNet50 [7] with cutmix as the baseline method to extract high-level features from the input images.

To evaluate the classification efficacy of the proposed network, OA (%) and Top-5 (%) are used as evaluation metrics, while Params (M) and Flops (G) are used to evaluate the amount of computation. Stochastic gradient descent (SGD) optimizer and batch normalization are applied as the regularizer in our experiments. All models are trained for 100 epochs on different datasets. Specifically, the learning rate is initially 0.05, which is multiplied by 0.1 at the 30th, 60th, and 90th epochs.

B. Datasets

1) FGSC-23: The FGSC-23 dataset is a high-resolution optical remote sensing dataset consisting of 4080 images primarily obtained from Google Earth and the Gaofen-1 (GF-1) Satellite. It contains 23 categories, including 22 categories of ships and a nonship category, and the sizes of the images range from 40 to 800 pixels. This dataset is designed for FGSC and is available for download at https://github.com/xiong577/ship-datasets.

2) FGSCR-42: The FGSCR-42 dataset is a publicly available collection of optical remote sensing images used for FGSC. The dataset comprises 9320 high-resolution images depicting 42 different types of ships. The images exhibit varying dimensions, ranging from 50 × 50 to approximately 500 × 1500. This dataset is intended to serve as a valuable resource for researchers and practitioners interested in developing and evaluating machine learning models for ship classification tasks. Its availability allows for the replication and comparison of results across various approaches, thereby enabling the advancement of the field toward more accurate and efficient algorithms.

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TABLE I
COMPARISONS OF CLASSIFICATION ACCURACIES UNDER DIFFERENT VALUES OF PARAMETER (λ) WITHOUT AL ON THE TWO DATASETS

Dataset	$\lambda = 2$	$\lambda = 1$	$\lambda = 0.5$	$\lambda = 0.1$
FGSC-23	83.63	84.49	**86.18**	84.97
FGSCR-42	94.91	95.64	**97.57**	96.70

TABLE II
COMPARISONS OF DIFFERENT CA MAP GENERATION METHODS IN MFCA ON FGSC-23 AND FGSCR-42

Methods	OA(%)	
	FGSC-23	FGSCR-42
Random	**85.33**	**95.65**
Reversed	84.97	93.34

C. Parameter Analysis

In this section, experiments are carried out to analyze the effects of λ on the performance of the proposed network. λ is an important parameter in the proposed network, which indicates the weight of the loss of MFCA and MCA in the total loss. As shown in Table I, the proposed network achieves the best performance when λ is set to 0.5.

D. Different Types of CAs

In this section, we compared two different CAs in our experiments. Randomly generated attention value for each location is sampled from a uniform distribution $U(0, 2)$. Reversed attention maps are obtained by subtracting the original attention from the maximum attention value for each sample. As shown in Table II, randomly generated attention performs better than reversed attention.

E. Ablation Studies

In this section, we conduct ablation experiments on two FGSC datasets, namely, FGSC-23 and FGSCR-42, to investigate the effectiveness of the proposed MFCA, MCA, and AL. The experimental results are presented in Table III. CA [22], contextual transformer attention (CoTA) [13], and weighted loss (WL) [12] are existing methods of spatial dimension, channel dimension, and loss function.

In the spatial dimension, CA is a counterfactual attention method to spatially mining feature causality. The proposed MFCA facilitates network learning of fine-grained features more effectively than CA by introducing mid-level features. MFCA enables the network to explore the causal relationship between mid-level features and classification results through mutual guidance and supervision among mid-level feature maps, CA maps, and high-level feature attention maps. The MFCA achieves competitive classification results even when images of certain classes exhibit relatively large differences from images of the same class. This is because MFCA enables the network to obtain a more discriminate feature representation, increasing the difference information between categories.

In the channel dimension, CoTA is a channel attention method capable of obtaining contextual channel feature maps while adding little computational effort. The MCA can effectively utilize the channel information of mid-level features so that each channel of the network can focus on a different area and reduce the impact of a single salient feature. Thus, MCA is able to focus on more fine-grained features than CoTA.

In the loss function, a simple summation of the loss functions leads to different weights of MFCA and MCA in the training, which leads to unsatisfactory results. WL is obtained by weighting the loss functions of each method, but due to the different magnitude of variation among the loss functions, this calculation can only roughly reconcile the loss functions. The proposed AL can dynamically adjust the weight of the loss function according to the training loss, which can effectively improve the training efficiency.

Specifically, the proposed MFCA and MCA improve the OA of the FGSC-23 dataset by 2.30% and 2.06% and improve the OA of the FGSCR-42 dataset by 3.24% and 2.08%. The proposed AL improves the OA of MFCA and MCA by 0.85% and 1.64% on FGSC-23 and FGSCR-42. The proposed method achieves better results than the existing methods. It is worth noting that the improvement resulting from MFCA is significantly higher on FGSC-23 than on FGSCR-42, which is attributed to the former having fewer categories and containing more different information between categories.

In summary, our method demonstrates their effectiveness in FGSC tasks. These methods improve the network's ability to learn fine-grained features, leading to a more discriminate feature representation and increased difference information between categories.

F. Comparisons With Other Methods

In this section, we conduct a comparative analysis of the proposed method against existing fine-grained classification methods, including ConvMixer [28], ConvNeXt [18], VAN [6], Inception-v3 [27], DenseNet [10], Bilinear-CNN [15], VGG16 [25], ResNet50 [7], ResNext-50 [38], AC-Net [11], and TASN [41]. The performance of the proposed method is evaluated against that of other methods on two datasets, as presented in Table IV.

ConvMixer also achieved outstanding results in fine-grained classification through convolution kernel parameter settings. VAN obtains a more complete attention relationship through decomposed large kernel convolution operations, but the obtained attention feature map is still insufficient for fine-grained classification problems. ConvNeXt achieved excellent results in FGSCR-42 relying on optimization strategies and network design, but the results in FGSC-23 were not satisfactory. This is because the image resolutions of the two datasets are quite different, resulting in the incompatibility of ConvNeXt's internal convolution parameter settings with FGSC-23. For the FGSC-23 dataset, the commonly used CNN models are Inception-v3 and DenseNet. On the other hand, Bilinear-CNN is specifically designed for remote sensing image classification tasks and fine-grained natural image recognition tasks. Similarly, for the FGSCR-42 dataset, general CNN models are used for comparisons, including VGG16 and DenseNet. Among them, Bilinear-CNN and TASN are fine-grained image recognition models. TASN extracts global features using the proposed trilinear attention module, leading to better classification results than other methods. In summary,

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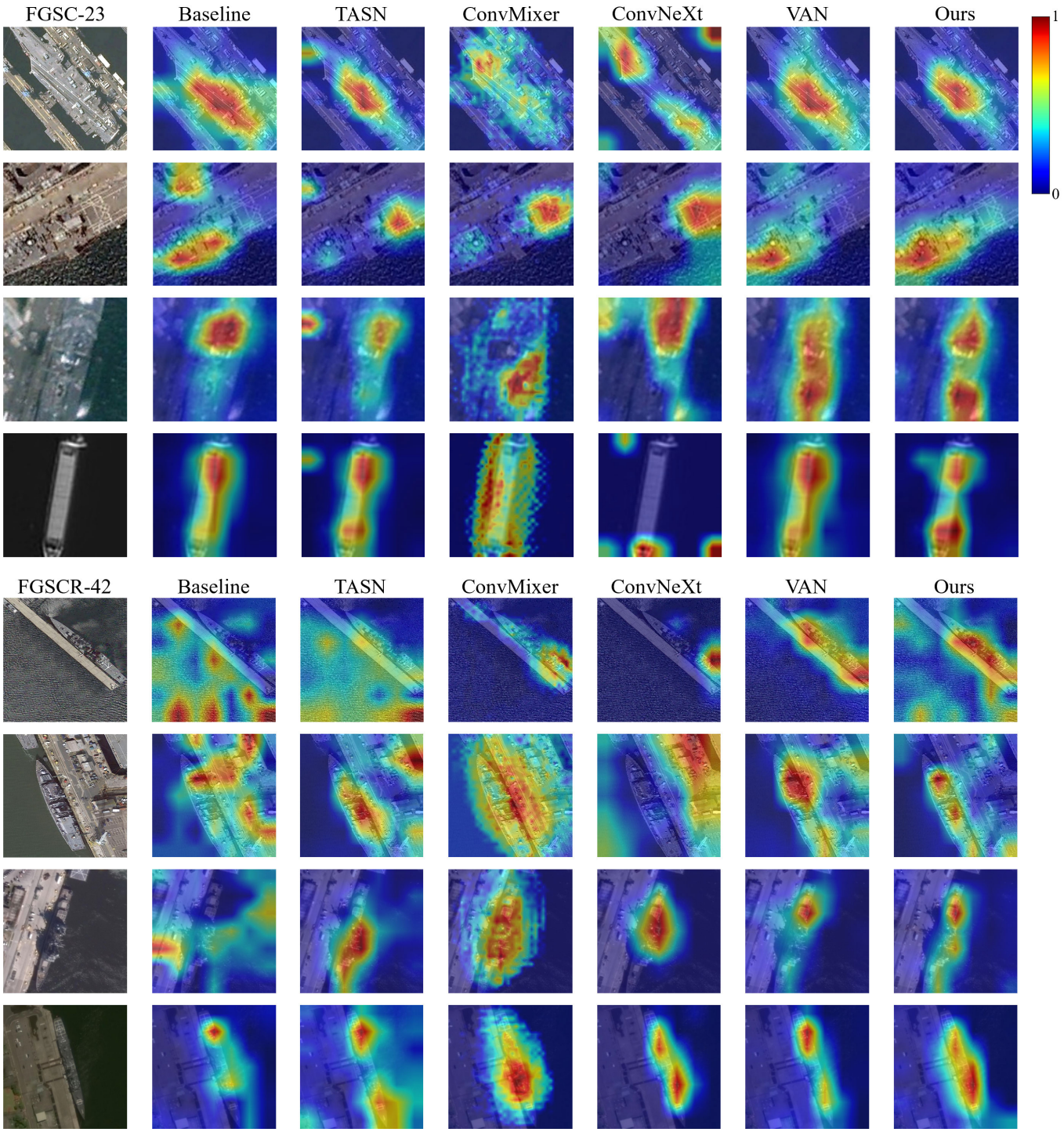


Fig. 8. Visualization of the attention maps with different methods on FGSC-23 and FGSCR-42 datasets.

our method improves FGSC-23 and FGSCR-42 by 3.15% and 5.16% compared to baseline and outperforms all listed methods, respectively.

The proposed method achieves state-of-the-art performance on both FGSC-23 and FGSCR-42 datasets. This further highlights that the introduction of mid-level features in the network effectively enhances its fine-grained classification performance, and the computational overhead associated with this method is negligible.

G. Visualization

In this section, we conduct visualization experiments to compare the attention maps generated by our model with those generated by the baseline model, TASN, ConvMixer, ConvNeXt, and VAN on the FGSC-23 and FGSCR-42 datasets. TASN and ConvNeXt are state-of-the-art on FGSC-23 and FGSCR-42, respectively. ConvMixer and VAN achieve excellent results on both datasets. Therefore, these four models and the baseline are chosen for comparison in the visualization

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TABLE III
RESULTS OF FGSC-23 AND FGSCR-42 DATASETS WITH DIFFERENT METHODS

Spitial		Channel		Loss		Params(M)	FLOPs(G)	OA(%)		Top-5(%)	
CA[22]	MFCA	CoTA[13]	MCA	WL[12]	AL			FGSC-23	FGSCR-42	FGSC-23	FGSCR-42
						24.96	4.12	83.03	92.41	97.56	99.30
✓						25.00	4.12	83.88	93.72	97.21	99.49
	✓					25.03	4.12	85.33	95.65	98.18	99.65
		✓				25.01	4.12	84.60	93.79	97.70	99.46
			✓			24.98	4.12	85.09	94.49	97.70	99.65
	✓		✓			25.05	4.12	85.21	95.10	97.58	99.62
	✓		✓	✓		25.05	4.12	85.70	95.77	97.94	99.04
	✓		✓		✓	25.05	4.12	**86.18**	**97.57**	**98.06**	**99.74**

TABLE IV
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ("–" REPRESENTS NO AVAILABLE EXPERIMENTAL RESULT)

Model	Params(M)	FLOPs(G)	OA(%)	
			FGSC-23	FGSCR-42
VGG16(2014) [25]	138.36	15.50	82.04	77.36
Bilinear-CNN(2015) [15]	-	-	84.00	89.53
ResNet50(2016) [7]	24.96	4.12	83.03	92.41
Inception-v3(2016) [27]	23.83	5.75	83.88	77.08
DenseNet(2017) [10]	20.01	4.37	84.00	88.69
TASN(2019) [41]	-	-	85.80	93.51
ResNext-50(2020) [38]	25.03	4.27	84.03	89.16
AC-Net(2020) [11]	-	-	69.80	88.40
Convmixer(2022) [28]	20.37	19.62	84.73	94.17
VAN(2022) [6]	26.58	5.03	84.00	94.94
Convnext(2022) [18]	49.47	8.69	81.45	95.58
ours	25.05	4.12	**86.18**	**97.57**

experiment. The visualization results are presented in Fig. 8. The results indicate that our method tends to focus on more than one region because our method effectively suppresses salient features and allows different feature channels to focus on different regions. This helps the network pay attention to more different subtle features, resulting in more attention responses. Moreover, the attention generated by our model focuses on the foreground better than other models, enabling the network to better recognize the objects.

For instance, the first and second rows of FGSC-23 and FGSCR-42 show that other comparison methods cause the network to pay too much attention to the background of the images, disturbing the learning process of fine-grained features. Our attention method focuses on objects while suppressing attention on the background, resulting in more accurate classification results. In addition, the third and fourth rows of FGSC-23 and FGSCR-42 show that other comparison methods tend to focus on a salient region of images. The classification process is based on the part feature with the highest response, which results in a loss of inconspicuous fine-grained feature information. According to the third row of FGSC-23 and the first row of FGSCR-42, TASN and ConvNeXt incorrectly focus on background information. According to the second row of FGSC-23 and the fourth row of FGSCR-42, ConvMixer tends to look for salient features, which makes it difficult to notice more diverse fine-grained features. VAN can focus on more diverse features than TASN, but it will focus on the background. Our attention method generates more responses for different parts while retaining the features with the highest responses, such as the third and fourth rows of FGSC-23 and the third and fourth rows of FGSCR-42. It can be seen that our method is more effective in capturing fine-grained part features.

V. CONCLUSION

In this article, we propose an adaptive mid-level attention network for FGSC in remote sensing images. Our network utilizes the MFCA to extract mid-level attention maps and causal attention maps from the input images, while the MCA further mines channel information from mid-level feature maps. Our approach employs causal inference to guide the learning of mid-level feature maps using raw feature maps and CA maps. In addition, we incorporate a discriminative module to mine discriminative features from channel information and a diversity module to focus the feature channels on more obvious feature regions. Finally, we propose AL that can adaptively adjust the weights of MFCA and MCA according to the training loss of ResNet50, thus better dynamically adjusting the training process of the network. Our method achieves state-of-the-art results compared to existing methods, demonstrating the effectiveness of our approach in FGSC.

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